UNIVERSITY OF DETROIT MERCY
COLLEGE OF ENGINEERING & SCIENCE

CSSE-2130 Java
Syllabus – Fall 2026
The University of Detroit Mercy
In Partnership with Detroit Green Technology Institute (DGTI)

Course Description

Java applications, Java applets, control structures, methods, arrays, strings and characters, object-oriented programming, graphics and Java 2D, basic graphical user interface components, exception handling, files and streams, and Java utilities.

Pre-requisites:

CSSE 1720 - Introduction to Programming II

Ability to program using the C++ programming language with an understanding of abstract data types.

Lecturer:

Ma'en Saleh, Ph.D.

Email: salehms3@udmercy.edu

Office hours

Days	China-time	Notes
Tuesday	4.00-5.00 PM	Send me an email to arrange meetings outside the regular schedule
Wednesday	1.00-2.00 PM	
Wendesday	4.00-5.00 PM	

Teaching Assistants:

1. MA Zhongbao email: 31601997@qq.com
2. YANG Long email: 562648787@qq.com

Class Meetings:

Format [modify as needed]: Online synchronous class using Blackboard collaborate. Attendance is mandatory.

Days	China-time	Notes
Monday	6.30-8.05 PM	Weeks 6-13 (7A-215)
Tuesday	2.00-3.35 PM	Weeks 6-13 (7A-215)
Thursday	2.00-3.35 PM	Weeks 6-13 (7A-215)

Course Objectives :

1. Design and develop programs using the object-oriented programming paradigm.
2. Apply features of object-oriented paradigm, such as modularity, abstraction, encapsulation, inheritance and polymorphism.
3. Analyze and solve problems from an object-oriented perspective.

Course Outcomes:

Upon completion of the course, students will be able to:

1. Understand the unique features of Java.
2. Perform data input and output (console and file).
3. Write and debug Java programs.
4. Create a graphical user interface.
5. Present final project clearly orally and in writing.

Program Learning Outcomes:

Graduates of this program will be able to:

1. Analyze a complex computing problem and apply principles of computing and other relevant disciplines to identify solutions.
2. Design, implement, and evaluate a computing-based solution to meet a given set of computing requirements in the context of the program's discipline.
3. Communicate effectively in a variety of professional contexts.
4. Recognize professional responsibilities and make informed judgments in computing practice based on legal and ethical principles.
5. Function effectively as a member or leader of a team engaged in activities appropriate to the program's discipline.
6. Apply computer science theory and software development fundamentals to produce computing-based solutions.

List of Topics:

1. Introduction to Java
2. Concepts of Object-oriented Programming.
3. Arrays, ArrayLists, and Exceptions.
4. Composition
5. Inheritance
6. Polymorphism, Abstract Classes, Interfaces Part I
7. Polymorphism, Abstract Classes, Interfaces Part II
8. GUI (JavaFX)

Recommended Textbook:

- **Java: How To Program, Early Objects.** Paul Deitel and Harvey Deitel. Pearson. 10th or 11th Edition.
- Tony Gaddis, and Godfrey Muganda (2018), **Starting Out with Java: From Control Structures through Objects**, Pearson, 4th Edition, ISBN: 978-0134787961.

- David J. Eck (2015), **Introduction to Programming Using Java**, 8th Edition, *Open Textbook*.

Tentative Course Schedule:

- **Week 1 (Oct. 5 – Oct. 11): Introduction to Java**
  - **Topics:**
    - **Overview of Java and its applications**
    - **Downloading and installing JDK and IDE**
    - **Writing and running the first Java program**
  - **Programming Practice: Basic Java syntax, compiling, and running programs**
  - **COs: 1, 3**

- **Week 2 (Oct. 12 – Oct. 18): Concepts of Object-Oriented Programming (OOP)**
  - **Topics:**
    - **Classes and objects**
    - **Encapsulation, constructors, and methods**
  - **Programming Practice: Implement small OOP programs using classes and methods**
  - **COs: 1, 3**

- **Week 3 (Oct. 19 – Oct. 25): Arrays, ArrayLists, Exceptions**
  - **Topics:**
    - **Single and multidimensional arrays**
    - **Using ArrayLists**
    - **Exception handling (try-catch-finally)**
  - **Programming Practice: Programs with arrays, ArrayLists, and exception handling**
  - **COs: 1, 2, 3**

- **Week 4 (Oct. 26 – Nov. 1): Composition + OOP Useful Methods + Strings + Characters + Mini Project 1**
  - **Topics:**
    - **"Has-a" relationships and designing classes with composition**
    - **Common OOP methods (toString, equals, compareTo)**
    - **String manipulation methods**
    - **Character class and utility methods**
  - **Programming Practice: Programs using composition, OOP methods, Strings, and Characters**
  - **Mini Project 1 Discussion/Due: OOP Concepts Project**
  - **COs: 1, 2, 3**

- **Week 5 (Nov. 2 – Nov. 8): Inheritance + Mini Project 2**
  - **Topics:**
    - **Extending classes**

        - **Method overriding and super keyword**
    - **Programming Practice: Implement inheritance hierarchies**
    - **Mini Project 2 Discussion/Due: Inheritance Project**
    - **Midterm Exam**
    - **COs: 1, 3, 5**

- **Week 6 (Nov. 9 – Nov. 15): Inheritance (continued) + Midterm Exam**
    - **Topics:**
        - **Advanced inheritance techniques**
    - **Programming Practice: Programs demonstrating inheritance**
    - **COs: 1, 3**

- **Week 7 (Nov. 16 – Nov. 22): Polymorphism, Abstract Classes, and Interfaces + Mini Project 3**
    - **Topics:**
        - **Polymorphism fundamentals**
        - **Abstract classes**
        - **Interfaces**
    - **Programming Practice: Programs with multiple interfaces and polymorphic behavior**
    - **Mini Project 3 Discussion/Due: Polymorphism/Interfaces Project**
    - **COs: 1, 3, 5**

- **Week 8 (Nov. 23 – Nov. 29): JavaFX and Scene Builder**
    - **Topics:**
        - **Integration of polymorphism and interfaces in larger programs**
        - **Introduction to JavaFX**
        - **JavaFX application structure: stages, scenes, and layouts**
        - **Designing user interfaces using Scene Builder**
        - **Connecting Scene Builder components to Java code**

    - **Programming Practice: Creating a simple JavaFX GUI using Scene Builder**
      Final Exam
    - **COs: 1, 3**

Note: Assignments and quizzes will be announced on Blackboard.

DGTI Policies[REQUIRED]:

Cell Phones Policy:

The use of cell phones (including voice, text, and any other applications not required for the course) is **not allowed** during lectures and other class activities. Cell phones must be **turned off or left at home** so as not to disturb the other students in the class. The consequences of using a cell phone or a cell phone ringing during class can include one or more of the following:

- The instructor will answer the call and/or share the communication with the entire class;

- The owner of the cell phone will be immediately asked to leave the classroom and may not return during that class period. Any work missed cannot be made up;
- The entire class will be given an immediate quiz; and
- The use or ringing of a cell phone during an exam will result in the termination of the exam at that point for the owner of the phone

* If you receive an emergency phone call, leave the classroom or laboratory and take the phone call in the hallway or outdoors.

Academic Expectations and Courtesy:

Students are expected to adhere to the following academic expectations and courtesy behavior:

- Students are expected to always conduct themselves in a **professional manner**. This expectation extends to **all** course related activities including the use of email;
- Email communications should be **professionally written;**
- Students must be courteous to their fellow classmates, teaching assistant, and professor. Talking during lectures is **not permitted**. If you are disturbing others in class, the professor will require you to leave; and
- Academic Policy requires students to submit any final grade concerns no later than 5 days after students have received official final grades from the administration. Specific dates are mentioned in the Grade Distribution section of the Syllabus.

Help with Assignments:

The professor and teaching assistants (if available for this course) are willing to help with the assignments in the class. However, students must demonstrate that they have made a good-faith effort at solving the problem before consulting with the professor or teaching assistants.

When asking questions, be prepared to demonstrate the efforts that you have made in solving the problem and be prepared to discuss your plan of attack. Statements such as "I don't understand the problem, please show me how to do it" will likely **not** be answered. You will be asked about what you do understand about the problem. Questions should be brought to the professor or teaching assistants well before the due date: Neither the professor nor the teaching assistant are obligated to make extensive homework consulting time available the day before the assignment due date. The professor and teaching assistants will make an effort to answer email and discussion board inquiries within 24 hours, but this is **not** guaranteed.

Grading for the course will be based primarily on, but may vary according to courses:

- **Attendance & class participation** (absences may lead to failure)
  - Students are required to **attend and be punctual** at every session.
- Timely submission of **homework /Assignments** (due dates)
  - The failure of a disk, a computer, or other storage media is not an acceptable excuse for late or missing homework assignments
- **Quizzes**
  - The professor reserves the right to give both announced and unannounced quizzes
- **Exams/ tests** (midterm and final exam)
  - Exam ownership: Faculty owns the exam and copying is strictly prohibited. If copying and/or photography is used, the student will automatically fail the class. You will be able to view completed exams but may not keep them or photograph them.
  - Legitimate excuse (i.e., illness, death in the family, etc.), you must contact the professor within 24 hours of the examination date (with proof, such as a doctor's note). Otherwise, you will receive a zero for that exam.
- Independent **project** /term project (with written reports and oral presentations)

Academic Integrity:

As members of an academic community engaged in the pursuit of truth and with a special concern for values, University of Detroit Mercy students must conform to the highest standard of honesty and integrity in their academic work. The fundamental assumption under which the University operates is that **work submitted by a student is a product of their own efforts**. Among the most serious academic offensives is **plagiarism**, submitting the ideas or work of another source without acknowledgment or documentation. The consequences of plagiarism or any act of academic dishonesty may range from failure on a course to dismissal from the university.

The professor takes academic integrity **very seriously**, and violations will be reported to the appropriate College authorities.

- All work submitted for this course must be the sole the work of the student or the team of students working on a group assignment.
- Any work (words, ideas, data, code, etc.) that is taken from another source must be properly cited and documented.

Violations of Academic Integrity:

Violations of academic integrity includes, but is not limited to the following:

1) Plagiarism:
   (a) Copying ideas or taking exact wording from published sources without using quotation marks and indicating where the words came from.
   (b) Paraphrasing from sources without citation.
   (c) Copying someone else's work and submitting it as your own.
   (d) Buying a paper or having someone else write it for you.

2) Inappropriate Collaboration:
   (a) Working together on individual tale-home exams.
   (b) Collaborating with students outside of the course on individual take-home exams.
   (c) Using student's work from previous years.
   (d) Handing in assignments (typically electronic submissions) where only superficial changes have been made.

3) Cheating:
   (a) Using materials not permitted by the instructor during exams or assignments.
   (b) Inappropriate copying answers from other students (on exams or assignments).
   (c) Altering graded exams or assignments and submitting them for regrading in order to get a better grade

4) Facilitating Academic Dishonesty:
   (a) Allowing another student to copy an assignment that was given as individual work.
   (b) Allowing another to copy exam answers.
   (c) Taking exams or completing assignments for another student.

Copyrights:

All course materials students receive or have online access to **are protected by copyright laws**. In addition, the textbook and the professor-provided resources are likewise **protected**. Students may use course materials

and make copies for their own use as needed, but unauthorized distribution and/or uploading of materials without the professor's express permission is strictly **prohibited**. Students who engage in the unauthorized distribution of copyrighted materials may be held in violation of the Student Code of Conduct, and/or liable under Federal and State laws. In addition, Copying, reposting, and submitting course materials (like completed essays, labs, homework, exams, quizzes, or other assignments ） to other websites or information repositories constitutes a violation of the Student Conduct policy and are **strictly prohibited**.

Recording of Class Sessions:

Video/audio recordings of class sessions (face-to-face or online) may be recorded (by the professor). Recordings will be shared via platforms with access limited to other members of the class.

Students may **not** record the professor without consent during lectures, laboratories, office hours, appointments, or discussions. The use of audio and/or video recording equipment or the making of audio recordings, video recordings, or transcriptions is not permitted. The use of all recording devices, including tape recorders, video cameras, still cameras, electronic voice recorders, computers, cell phones, tablets, and any other recording devices is not permitted.

Grading Policy:

Assignments	15%
Mini-Projects	15%
Quizzes	10%
Attendance (Extra)	5%
Midterm Exam	25%
Final Exam	30%

Learning Activities

To achieve the objectives of the course, students will carry out several learning activities:

1. **Readings**: The students are expected to read the assigned textbook chapter, slides, online resources, videos, and tutorial materials.
2. **Lectures**: Students are expected to attend every lecture; this is where the course material will be discussed and ambiguities clarified. Class participation is highly encouraged. The technologies to be applied in the project and the assignments will be presented in the lectures via examples and demos. Practical demos are often done in class including designing and coding and end-to-end solution to a given problem.
3. **Assignments**: Those includes Reading assignments, Programming assignments and In class assignments. They will be given so that students practice and apply the material covered in class. Each assignment will require the students to practice the material learned during the course. All assignments should be submitted on time. Late assignments will be graded out of 75% up to 3 days.
4. **Mini-Projects**: Students will develop small-scale software applications individually, applying key Java concepts such as object-oriented programming, collections, multithreading, and GUI development. These mini-projects are designed to enhance technical skills, problem-solving ability, and practical application, preparing students for real-world software development.

   Grading Criteria:

   - Functionality: 40% – The project should work correctly and meet the requirements.

- Design: 25% – The project should be well-structured and demonstrate good software design principles.
- Effort & Implementation: 15% – The project should reflect a solid individual effort and proper implementation.
- Documentation: 20% – Clear and complete documentation of the project.

5. **Exams:** To keep the students on track continuous assessment is adopted in-class using quizzes and off-class using lab assignments. The midterm has a theoretical part and the final exams have both a theoretical part and a practical programming part to build a solution to a problem. Quizzes are used for continued assessment.

NOTE: Students with questions related to their course grade or achievement for FALL 2026 must contact instructors by Monday, November 30th, 2026 at 1:00 PM China-Time. Any grade inquiry beyond specified date might not be addressed.

Grading Scale:

Percentage	Letter Grade
93 – 100%	A
90 - 92%	A^-
87 - 89%	B^+
83 – 86%	B
80 – 82%	B^-
77 – 79%	C^+
73 – 76%	C
70 – 72%	C^-
65 – 69%	D^+
60 – 64%	D
< 60%	F

Important Dates:

Note: *The instructor reserves the right to change the outline to accommodate student pace in understanding of the subject matter.*

Follow Blackboard Weekly activities as posted on BB class site